



Longwave thermosolutal instability in a moderately dense nanoparticle suspension

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Abstract We carry out the long-wave analysis of the monotonic thermo-solutocapillary instability coupled with the Soret effect in a layer of a moderately dense nanofluid layer with the deformable interface exposed to the ambient gas phase in the gravity field. The results of the long-wave analysis are compared with those obtained from the fully numerical solution of the pertinent linear eigenvalue problem formulated for the problem at hand, and the agreement is found to be excellent.

1 Introduction

A quiescent horizontal thin liquid layer subjected to heat and mass transfer can become unstable and develop different convective states. The physical mechanisms for the onset of instabilities are categorized into (i) buoyancy-driven instability and (ii) surface-tension-driven instability. The former takes place due to the adverse gradient of temperature-concentration-dependent density. However, the latter occurs due to the surface tension gradient at the liquid-gas interface which can be driven by inhomogeneities in local temperature, solute concentrations or other relevant factors. Surface-tension-driven instabilities are ubiquitous in nature and in many industrial applications [1–3]. Thermocapillary instabilities prevail in a microgravity environment and in sufficiently thin liquid layers [4–7].

Pearson [8] performed the linear stability analysis in a layer of a Newtonian fluid with a nondeformable interface in two different cases, namely that of the constant bottom temperature and that of a specified heat flux at the supporting substrate. He was motivated by an attempt to understand instability issues faced in paint coating applications. Pearson found the threshold values for the thermocapillary instability in terms of the Marangoni number, and that it may be either long-wave or finite-wave. Subsequently, Scriven and Sternling [9] reported a significant contribution of the interfacial deflection to the stability boundary. Smith [10] further corrected their result and showed the stabilization of the long-wave deformational mode.

In liquid mixtures such as binary mixtures, the temperature gradient induces the gradient of solute mass flux via what is known as the Soret effect [11, 12]. Joo [13] and Oron and Nepomnyashchy [14] found the onset of monotonic and oscillatory thermocapillary instability in the settings of layers with either deformable or nondeformable interface. They found that the oscillatory instability emerges due to the competition between the destabilizing thermocapillarity and stabilizing solutocapillarity when the binary mixture is heated at the substrate. Podolny et al. [15] studied the linear stability of a binary mixture with the Soret effect in the long-wave limit. They characterized two distinct long-wave scaling laws relating the critical wavenumber k_c and the Biot number B , which represents the ratio between the heat flux to the atmosphere and the heat flux due to conduction, namely, $k_c \sim B^{1/4}$ and $k_c \sim B^{1/2}$.

Nanofluids represent a colloidal dispersion made of a mixture of a base fluid and nanoparticles of diameter d_p^* between 1 and 100 nm. The thermophysical properties of a nanofluid such as density, kinematic viscosity, thermal diffusivity, and Soret diffusion coefficient are known to depend on the local bulk concentration of nanoparticles

The paper is dedicated to the memory of Professor Etienne Guyon.

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[16]. The viscosity of a nanofluid increases with the local particle concentration and can be approximated via a semi-empirical model [17, 18]. The Soret effect is also very important in the dynamical properties of nanofluids, and the Soret coefficient may be proportional to the local particle concentration [19]. The thermal conductivity of nanofluids also depends on the particle concentration and several types of this dependence are given by Buongiorno [16].

Gandhi et al. [20] have recently introduced the theoretical framework for fluid mechanics of moderately dense nanofluids where all of their thermophysical properties depend on the local particle concentration. One of the unexpected properties of the thermo-solutocapillary instability considered there was the impact of the thermal conductivity variation of the nanofluid on the type of the instability emerging in the system. This effect turns to be another example of the crucial role of the stratification of the thermal diffusivity on the emergence of instability in a two-layer Newtonian fluid system [21, 22].

The purpose of this paper is to carry out the long-wave linear stability analysis in the distinguished asymptotic limit and to validate its results with those of the fully numerical investigation of the problem reported in [20].

The plan of the paper is as follows. Section 2 presents the problem formulation and a set of governing equations and boundary conditions based on thermophysical properties of the nanofluid that depend on local particle concentration. Also, this section introduces the base state whose linear stability is addressed in this paper. Section 3 displays the full eigenvalue problem that determines the stability properties of the system at hand. Section 4 presents a justification for a simplifying assumption of “incompressibility” for the continuity equation which greatly simplifies also the long-wave stability analysis given in Sect. 5. Finally, Sect. 6 summarizes the findings of the paper.

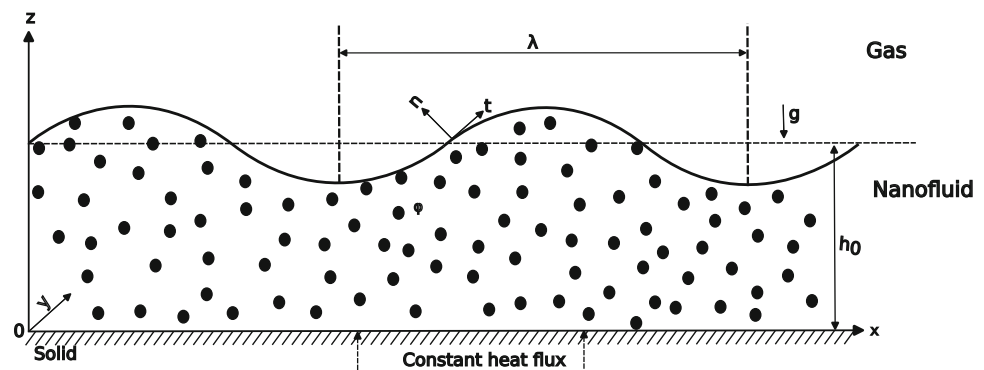
2 Problem formulation and governing equations

We consider a two-dimensional (2D) nanofluid layer of a mean thickness h_0^* , density ρ^* , dynamic viscosity μ^* , kinematic viscosity $\nu^* = \mu^*/\rho^*$, thermal conductivity K^* , heat capacity c^* , and thermal diffusivity $\kappa^* = K^*/\rho^*c^*$. In what follows, we denote the properties of the nanofluid, the base fluid, and the nanoparticles with subscripts nf , bf , and np , respectively. The nanofluid layer of thickness h^* resting on a solid planar substrate is exposed to the quiescent gas environment held at constant pressure p_∞^* and temperature T_∞^* with the deformable liquid-gas interface in the gravity field g^* . The frame of reference is chosen so that the x^* axis is located on the substrate, whereas the axis z^* is normal to the axis x^* and directed into the fluid layer opposite to the direction of gravity, hence the substrate and the deformable liquid-gas interface are located at $z^* = 0$ and $z^* = h^*$, respectively, as illustrated in Fig. 1.

The fluid is assumed to be a moderately dense nanoliquid mixture of a Newtonian base fluid with nanoparticles of diameter $d_p^* \approx 10$ nm whose volumetric concentration $\phi^* = \phi^*(x^*, z^*, t^*)$ is varying in time t^* and space. The entire system is subjected to the prescribed heat flux Qq^* with $Q = \pm 1$ and $q^* > 0$ at its solid bottom in the direction normal to the latter. Note that the value of Q is related to the direction of the heat flux, so that the cases of $Q = 1$ and $Q = -1$ correspond to heating and cooling at the solid-liquid interface, respectively. In what follows, we mainly concentrate on the case of heating at the substrate, $Q = 1$. An imposed heat flux leads to the emergence of the temperature field within the layer varying with time and space $T^* = T^*(x^*, z^*, t^*)$.

Before we formulate an appropriate set of governing equations and boundary conditions we comment on the thermophysical properties of moderately dilute nanosuspensions. In the case of a dilute binary mixture, its thermophysical properties such as density, heat capacity, viscosity, and thermal conductivity are normally assumed to be constant, and this assumption provides a fair approximation. However, in the case of moderately dilute mixtures, the thermophysical properties of the nanofluid depend on the particle concentration [16].

Fig. 1 Nanofluid (nanoparticles diameter $d_p^* = 10\text{--}100$ nm) layer on the solid substrate subjected to a constant heat flux at the substrate and exposed to the gas phase at its deformable interface



The density and the specific heat capacity of the nanofluid are determined as the weighted sum of the respective properties of the base fluid and the nanoparticles

$$\rho_{nf}^*(\phi^*) = (1 - \phi^*)\rho_{bf}^* + \phi^*\rho_{np}^*, \quad c_{nf}^*(\phi^*) = \frac{(1 - \phi^*)(\rho^*c^*)_{bf} + \phi^*(\rho^*c^*)_{np}}{\rho_{nf}^*}. \quad (1a)$$

In the case of nanofluid viscosity, we make use of the expression suggested by [17] for a concentrated suspension given by

$$\mu_{nf}^*(\phi^*) = \mu_{bf}^* \left(1 - \frac{\phi^*}{\phi_m}\right)^{-2}, \quad (1b)$$

where ϕ_m is the maximal packing fraction taken in our computations as $\phi_m = 0.65$ related to the random close packing (RCP) volume fraction of nanoparticles. This model illustrates the power-law variation of the viscosity with the nanoparticle concentration diverging when the nanoparticle concentration becomes too high, $\phi^* \rightarrow \phi_m$.

The thermal conductivity of nanofluids is also found to depend on the particle concentration, and is suggested for a nanofluid containing alumina Al_2O_3 nanoparticles in water by Buongiorno [16] to be

$$K_{nf}^*(\phi^*) = K_{bf}^*(1 + a\phi^*), \quad (2)$$

where a is a constant describing the inhomogeneity of the thermal conductivity, $a = 7.47$.

From among the seven different physical mechanisms relevant for nanoparticle motion in a nanofluid [16], we consider two dominant ones, namely, Brownian diffusion and the Soret effect known also as thermophoresis in a differentially heated fluid. The Brownian diffusion coefficient is given by the Stokes-Einstein formula [23]

$$D_B = \frac{K_B T^*}{3\pi\mu_{bf}^* d_p^*}. \quad (3)$$

where $K_B = 1.380649 \times 10^{-23}$ J/K is the Boltzmann constant.

The second mechanism is due to the fact that the nanofluid layer is non-isothermal. The temperature gradient imposed on a system induces a flux of nanoparticles, and this phenomenon is known as the thermophoresis or the Soret effect. The Soret or thermal diffusion coefficient in a nanofluid is proportional to the particle concentration ϕ^* and is expressed by [24, 25]

$$D_T = \left(\frac{0.26K_{bf}^*}{2K_{bf}^* + K_{np}^*} \right) \frac{\mu_{bf}^*}{\rho_{bf}^*} \phi^*, \quad (4)$$

Here, we emphasize that the kinematic viscosity, thermal diffusivity, and the Soret effect of a colloidal suspension strongly depend on the local concentration of the particles [26]. In what follows, we consider the total nanoparticle mass flux as a combination of the Brownian and Soret diffusions via

$$\mathbf{j}_p = -\rho_p^* \left(C_B T^* \nabla^* \phi^* + C_T \phi^* \frac{\nabla^* T^*}{T^*} \right), \quad (5)$$

where $C_B = D_B/T^*$, $C_T = D_T/\phi^*$ and $\nabla^* = (\partial_{x^*}, \partial_{z^*})$ with subscripts denoting partial differentiation with respect to the corresponding variable.

The heat flux in the nanofluid is given by Fourier's law of heat conduction

$$\mathbf{j}_T = -K_{nf}^* \nabla^* T^*. \quad (6)$$

The Dufour effect is exceedingly weak in liquid mixtures [11, 14, 27] and can be safely neglected in the case at hand.

The surface tension at the liquid-gas interface is assumed to depend on both the interfacial temperature and nanoparticle concentration, and for sufficiently small variations of temperature and particle concentration at the interface, it is adequately approximated by a linear function

$$\sigma^*(T^*, \phi^*) = \sigma_r^* - \sigma_{T^*}^*(T^* - T_r^*) - \sigma_{\phi^*}^*(\phi^* - \phi_r^*), \quad (7a)$$

where

$$\sigma_{T^*}^* = -\partial\sigma^*/\partial T^* > 0 \text{ and } \sigma_{\phi^*}^* = -\partial\sigma^*/\partial\phi^* > 0. \quad (7b)$$

where σ_r^* is the reference value of the surface tension, and both $\sigma_{T^*}^*$ and $\sigma_{\phi^*}^*$ are positive, thus, the surface tension linearly decreases with the temperature and particle concentration at the interface. The alumina nanoparticle dispersion in distilled water exhibits a vanishing value of the adsorption/desorption coefficient ratio, hence, it is possible to neglect the interfacial kinetics mechanism in this situation. However, we note that the nanoparticle dispersion in different liquids such as non-stabilized water, n-decane, n-dodecane, n-hexadecane, etc., exhibits a significant contribution of the interfacial kinetics. In those cases, the consideration of the interfacial kinetics becomes necessary [28] and instead of the interfacial value of the bulk particle concentration ϕ^* , a surface concentration Γ^* needs to be used in Eqs. (7a) and (7b).

In terms of dimensionless variables defined via

$$\begin{aligned} (x^*, z^*) &= h_0^*(x, z), \quad h^*(x^*, t^*) = h_0^*h(x, t), \quad t^* = \frac{h_0^{*2}}{\nu_{bf}^*}t, \quad \mathbf{u}^* = \frac{\kappa_{bf}^*}{h_0^*}\mathbf{u}, \\ p^* &= p_\infty^* + \frac{\mu_{bf}^*\kappa_{bf}^*}{h_0^{*2}}p, \quad T^* = T_\infty^* + \frac{q^*h_0^*}{K_{bf}^*}T, \quad \phi^* = \phi_m\phi, \\ K_{nf}^* &= K_{bf}^*K_{nf}, \quad \rho_{nf}^* = \rho_{bf}^*\rho_{nf}, \quad \mu_{nf}^* = \mu_{bf}^*\mu_{nf}, \quad (\rho^*c^*)_{nf} = (\rho^*c^*)_{bf}(\rho c)_{nf}, \end{aligned} \quad (8)$$

where $\mathbf{u} = (u, w)$ is the flow field, dimensional variables are decorated with an asterisk and the corresponding dimensionless variables are denoted with the same symbol but without the asterisk, we now present a set of nondimensional governing equations and boundary conditions [20]:

$$P(\rho_{nf})_t + \nabla \cdot (\rho_{nf}\mathbf{u}) = 0, \quad (9a)$$

$$(\rho_{nf}\mathbf{u})_t + P^{-1}\nabla \cdot (\rho_{nf}\mathbf{u}\mathbf{u}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} - \rho_{nf}G\mathbf{e}_z, \quad (9b)$$

$$(\rho c)_{nf}[PT_t + (\mathbf{u} \cdot \nabla)T] = \nabla \cdot (K_{nf}\nabla T), \quad (9c)$$

$$PL^{-1}\phi_t + L^{-1}\nabla \cdot (\mathbf{u}\phi) = \nabla^2\phi + \nabla \cdot (\eta\phi\nabla T), \quad (9d)$$

with $\nabla \equiv (\partial_x, \partial_z)$, $\boldsymbol{\tau}$ is the deviatoric part of the stress tensor, namely,

$$\boldsymbol{\tau} = \mu_{nf}(\nabla\mathbf{u} + \nabla\mathbf{u}^\dagger) - \frac{2}{3}\mu_{nf}(\nabla \cdot \mathbf{u})$$

with the superscript $\dagger$ denoting transpose of the corresponding tensor. We note that Stokes' hypothesis [29] of a zero value for the dilatational viscosity is used here.

The boundary conditions at $z = 0$:

$$\mathbf{u} = 0, \quad K_{nf}T_z = -\mathcal{Q}, \quad \phi_z + \eta\phi T_z = 0. \quad (10a)$$

The boundary conditions at $z = h$:

$$Ph_t + uh_x = w, \quad (10b)$$

$$-p - \frac{2\mu_{nf}}{3(1+h_x^2)}\left\{h_x(-2h_xu_x + 3w_x) + 3h_xu_z + (h_x^2 - 2)w_z + u_x\right\} = \frac{\Sigma_0h_{xx}}{(1+h_x^2)^{3/2}}, \quad (10c)$$

$$\begin{aligned} &\mu_{nf}[2h_x(u_x - w_z) + (h_x^2 - 1)(u_z + w_x)] \\ &= \left[M_T(T_x + h_xT_z) + M_S(\phi_x + h_x\phi_z)\right]\sqrt{1+h_x^2}, \end{aligned} \quad (10d)$$

$$K_{nf}(T_z - h_x T_x) + BT\sqrt{1 + h_x^2} = 0, \quad \phi_z - h_x \phi_x + \eta \phi(T_z - h_x T_x) = 0. \quad (10e)$$

Finally, the average particle concentration in the layer is specified by

$$\frac{1}{\ell} \int_0^1 \left(\int_0^\ell \phi(x, z, t) dx \right) dz = \Phi \quad (11)$$

with ℓ being a lateral extent of the layer for all times t and a fixed prescribed positive value Φ . Here P , G , L , B , M_T , M_S , Σ_0 , η , and Q are, respectively, the Prandtl, the modified Galileo, Lewis, Biot, thermal Marangoni, solutal Marangoni, dimensionless surface tension, Soret coefficient, and direction of the applied heat flux,

$$\begin{aligned} P &= \frac{\nu_{bf}^* \rho_{bf}^* c_{bf}^*}{K_{bf}^*}, \quad G = \frac{g^* h_0^3}{\nu_{bf}^* \kappa_{bf}^*}, \quad L = \frac{C_B T_\infty^* \rho_{bf}^* c_{bf}^*}{K_{bf}^*}, \quad B = \frac{\hat{q} h_0^*}{K_{bf}^*}, \quad M_T = \frac{q^* h_0^{*2} \sigma_{T^*}^*}{\mu_{bf}^* K_{bf}^* \kappa_{bf}^*}, \\ M_S &= \frac{\phi_m h_0^* \sigma_{\phi^*}^*}{\mu_{bf}^* \kappa_{bf}^*}, \quad \Sigma_0 = \frac{\sigma_r^* h_0^*}{\mu_{bf}^* \kappa_{bf}^*}, \quad \eta = \frac{C_T q^* h_0^*}{C_B T_\infty^{*2} K_{bf}^*}, \quad Q = \pm 1. \end{aligned} \quad (12a)$$

Here, $\hat{q}$ is the rate of heat transfer from the liquid phase to the gas phase by convection according to Newton's law of cooling. In addition, the dimensionless form of the thermophysical properties appearing in the set of Eqs. (9) and (10) is

$$\begin{aligned} \rho_{nf}(\phi) &= (1 - \phi \phi_m) + \phi \phi_m \rho_{np}, \quad (\rho c)_{nf}(\phi) = (1 - \phi \phi_m) + \phi \phi_m (\rho c)_{np}, \\ K_{nf}(\phi) &= 1 + a \phi \phi_m, \quad \mu_{nf}(\phi) = (1 - \phi)^{-2}. \end{aligned} \quad (13)$$

Equations (9a)–(9d) represent the continuity, the momentum conservation, the energy conservation, and the nanoparticle mass transfer equations, respectively. The boundary conditions (10a) represent, respectively, no-slip – no-penetration, a constant prescribed heat flux, and a zero total mass flux at the solid substrate supporting the fluid layer from beneath. At the deformable interface of the layer $z = h$, the boundary conditions (10b)–(10d) express the kinematic boundary condition, and the continuity of the normal and tangential stress boundary conditions, respectively. Finally, Eq. (10e) presents the heat transfer boundary condition and a zero total mass flux at the layer interface.

We now determine a quiescent ($\mathbf{u} = \mathbf{u}_0 = \mathbf{0}$) steady base state of the system with the flat interface $h = 1$ and with the temperature and concentration fields depending solely on z , $T = T_0(z)$, $\phi = \phi_0(z)$, of the problem given by Eqs. (9)–(10). It is determined by

$$p_{0,z} = -((1 - \phi_0 \phi_m) + \phi_0 \phi_m \rho_{np})G, \quad (14a)$$

$$[(1 + a \phi_m \phi_0)T_{0,z}]_z = 0, \quad (14b)$$

$$(\phi_{0,z} + \eta \phi_0 T_{0,z})_z = 0, \quad (14c)$$

$$z = 0: \quad (1 + a \phi_m \phi_0)T_{0,z} = -Q, \quad \phi_{0,z} + \eta \phi_0 T_{0,z} = 0, \quad (15a)$$

$$z = 1: \quad p_0 = 0, \quad (1 + a \phi_m \phi_0)T_{0,z} + BT_0 = 0, \quad \phi_{0,z} + \eta \phi_0 T_{0,z} = 0. \quad (15b)$$

Finally, following Eq. (11)

$$\int_0^1 \phi_0(z) dz = \Phi. \quad (16)$$

The solution of the problem Eqs. (14)–(15) for the pressure, temperature, and concentration components is found to be

$$p_0 = \frac{1}{2\eta Q a \phi_m} \left\{ G \phi_m (\rho_{np} - 1) \left[\mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \eta Q)) \right. \right. \\ \left. \left. - \mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \eta Q z)) \right] \times \left[\mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \eta Q)) \right. \right. \\ \left. \left. + \mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \eta Q z)) + 2 \right] \right\} + G(1 - z), \quad (17a)$$

$$T_0 = \frac{1}{\eta B} \left[B \ln(\mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \eta Q))) \right. \\ \left. - B \ln(\mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \eta Q z))) + \eta Q \right], \quad (17b)$$

$$\phi_0 = \frac{1}{a \phi_m} \mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \eta Q z)), \quad (17c)$$

where $\mathcal{W}(z)$ is the principal (increasing) branch of the Lambert W function [30] defined via

$$z = \mathcal{W}(z) \exp \mathcal{W}(z) \quad (18)$$

and $\gamma(\Phi)$ is determined by substitution of $\phi_0(z)$ from Eq. (17c) into Eq. (16). We note that the secondary (decreasing) branch of the Lambert W function yields negative values of Φ and is therefore irrelevant.

Figure 2 presents several examples of the base state in terms of the temperature $T_0(z)$ (the right panel) and particle concentration $\phi_0(z)$ (the left panel) in the case of heating at the substrate, $Q = 1$, for various values of the average particle concentration Φ and the temperature drop across the layer in the base state ΔT^* determined via

$$\Delta T^* = q^* h_0^* / K_{bf}^*. \quad (19)$$

We note that the concentration profiles exhibit an unstable stratification, owing to the positive value of the Soret coefficient η . We also observe that the concentration profiles display a nonlinear variation across the layer thickness z for higher values of the temperature difference ΔT^* , e.g., $\Delta T^* = 10K$.

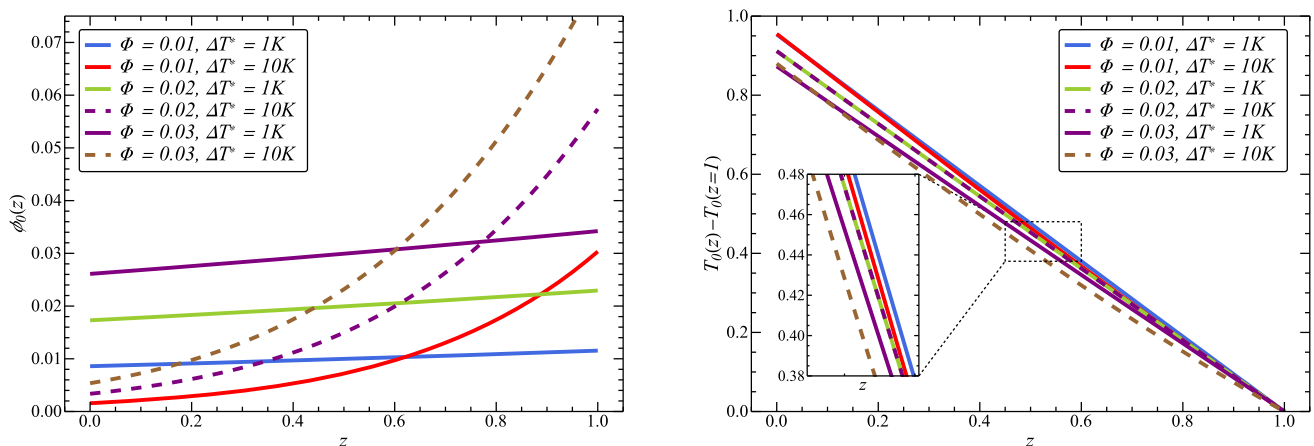


Fig. 2 Variation of the quiescent base-state concentration $\phi_0(z)$ (left panel) and temperature presented as $T_0(z) - T_0(z = 1)$ (right panel) with height z , for the case of heating at the substrate $Q = 1$, and different values of the averaged bulk concentration Φ and different values of the temperature difference ΔT^* with $B = 0.01$, $\eta = 0.31$, and $a = 7.47$. The solid (—) and dashed (---) lines represent the data corresponding to temperature difference $\Delta T^* = 1K$ and $\Delta T^* = 10K$ across the nanofluid layer

3 Linear stability analysis

We now investigate the linear stability of the base state given by Eq. (17). To do this, the dependent variable fields are linearized around the base state in the form

$$\mathbf{u} = \mathbf{u}_0 + \bar{\mathbf{u}} = \bar{\mathbf{u}}, \quad p = p_0 + \bar{p}, \quad T = T_0 + \bar{T}, \quad \phi = \phi_0 + \bar{\phi}, \quad h = h_0 + \bar{\zeta}, \quad (20)$$

where the values with an overbar represent the disturbances of the corresponding fields. Furthermore, we linearize the thermophysical properties along the dimensionless base state concentration ϕ_0 , i.e.,

$$\left. \begin{aligned} \rho_{nf} &= R_0 + \bar{R} = (\phi_m(\rho_{np} - 1)\phi_0 + 1) + (\phi_m(\rho_{np} - 1)\bar{\phi}), \\ (\rho c)_{nf} &= H_0 + \bar{H} = (\phi_m((\rho c)_{np} - 1)\phi_0 + 1) + (\phi_m((\rho c)_{np} - 1)\bar{\phi}), \\ K_{nf} &= K_0 + \bar{K} = (a\phi_m\phi_0 + 1) + \bar{\phi}a\phi_m, \\ \mu_{nf} &= M_0 + \bar{M} = \frac{1}{(\phi_0 - 1)^2} - \frac{2\bar{\phi}}{(\phi_0 - 1)^3}, \end{aligned} \right\} \quad (21)$$

where for brevity the symbols $\bar{X} \equiv (\bar{R}, \bar{M}, \bar{K}, \bar{H})$ stand for disturbances of ρ_{nf} , μ_{nf} , K_{nf} and $(\rho c)_{nf}$, respectively, represented each by the second term in the corresponding equation in (21). The values of $\bar{R}_0$, $\bar{M}_0$, $\bar{K}_0$ and $\bar{H}_0$ are given by the first term in the corresponding equation in (21).

In what follows, we constrain our study to the case of two-dimensional perturbations in the $x - z$ plane. Accordingly, the 2D equations are rewritten in terms of the vertical velocity component $\bar{w}$ of the flow field perturbations by satisfying the equation of mass conservation. By applying the $\nabla \times$ operator to the momentum conservation equations and differentiating the normal stress balance boundary condition along the interface followed by substituting there the pressure gradient components obtained from the momentum conservation equations, we completely eliminate the pressure from the momentum conservation equations.

Normal perturbations modes are introduced in the form

$$(\bar{u}, \bar{w}, \bar{T}, \bar{\phi}, \bar{\zeta}, \bar{X})(x, z, t) = (u(z), w(z), \theta(z), \varphi(z), \zeta, X) \exp(ikx + \lambda t) \quad (22)$$

where u , w , θ , φ , ζ , and X are the amplitudes of the horizontal and vertical fluid velocity components, temperature, concentration, interfacial deformation, and thermophysical properties perturbations, respectively, and k and λ are the wavenumber and the growth rate of the disturbances, respectively. In this representation, positive and negative values of the real part of λ correspond to instability and stability regimes of the system, respectively.

The eigenvalue problem which determines the growth rate λ of the disturbance as a function of the wavenumber k and the problem parameters reads

$$R_0(w' + iku) + \lambda PR + wR'_0 = 0, \quad (23a)$$

$$-iGkR + k^2M_0u' + 2k^2uM'_0 - ikw(k^2M_0 + \lambda R_0 + M''_0) + ikM_0w'' - u'''M_0 - 2u''M'_0 + \lambda R_0u' - u'M''_0 + \lambda uR'_0 = 0, \quad (23b)$$

$$-T''_0K + \theta(k^2K_0 + \lambda PH_0) + T'_0(wH_0 - K') - K_0\theta'' - K'_0\theta' = 0, \quad (23c)$$

$$\varphi(k^2L - \eta LT''_0 + \lambda P) + \eta k^2L\theta\phi_0 + \phi_0(iku - \eta L\theta'' + w') + \phi'_0(w - \eta L\theta') - \eta LT'_0\varphi' - L\varphi'' = 0. \quad (23d)$$

The boundary conditions at $z = 0$ are

$$\mathbf{u} = 0, \quad \theta'(a\phi_m\phi_0 + 1) + a\phi_m\varphi T'_0 = 0, \quad \eta\phi_0\theta' + \eta\varphi T'_0 + \varphi' = 0. \quad (23e)$$

The boundary conditions at the equilibrium c interface $z = 1$ are

$$\zeta\lambda P - w = 0, \quad (23f)$$

$$T'_0(a\phi_m(\zeta\phi'_0 + \varphi) + B\zeta) + (a\phi_m\phi_0 + 1)\theta' + \zeta T''_0(a\phi_m\phi_0 + 1) + B\theta = 0, \quad (23g)$$

$$\zeta\phi_0'' + \eta\phi_0(\theta' + \zeta T_0'') + \eta T_0'(\zeta\phi_0' + \varphi) + \varphi' = 0, \quad (23h)$$

$$kM_S(\zeta\phi_0' + \varphi) + kM_T(\zeta T_0' + \theta) + M_0(kw - iu') = 0, \quad (23i)$$

$$i\zeta k^3 \Sigma_0 + u(2k^2 M_0 + \lambda R_0) + i(-\zeta k p_0' + kM_0 w' - kM_0' w + iM_0 u'' + iM_0' u') = 0. \quad (23j)$$

Here, thermophysical properties with and without zero subscripts depict the thermophysical properties with the base state nanoparticle concentration and their perturbations due to perturbed nanoparticle concentration, respectively (see Eq. (21)). We finally note that normal perturbations imposed on the bulk concentration ϕ preserve the average bulk concentration due to periodicity of the x -component of the perturbation $\bar{\phi}$.

To precondition the eigenvalue problem for a more efficient numerical solution, we substitute the expression of the perturbed longitudinal velocity component u obtained from Eq. (23a) into the rest of the equations of the eigenvalue problem (23). We also extract the expression for the interfacial deformation ζ from Eq. (23j) and substitute it into the rest of the boundary conditions at $z = 1$. Hence, our eigenvalue problem is formulated in terms of w , θ , φ and in this format it will be treated numerically. We also note here that in the case of the non-deformable interface the pressure will be eliminated without substituting its gradients into the normal stress boundary condition since the latter is obsolete, and $\zeta = 0$ will be imposed.

The critical values for the Marangoni number corresponding to the onset of the system instability are obtained from the numerical solution of the linear eigenvalue problem (EVP) given by Eq. (23) or any of its simplified versions. The numerical analysis is carried out using the open-source code developed by Pearce et al. [31]. It is based on the shooting method and numerical construction of the complex valued Evans function [32] whose roots represent the eigenvalues of the given EVP. More details related to the numerical approach may be found in [20].

4 An incompressibility simplification

As follows from the problem statement, the fluid density depends on the local particle concentration which in turn evolves in both time and space. Therefore, in general, the approach to the problem analysis should consider the fluid as compressible. However, we find that despite that the effect of compressibility is weak in most parameter regimes we have considered, and therefore may be neglected. This fact leads to significant simplification of the derivations presented below in Sect. 5.

It is possible to rewrite the non-dimensional form of the nanofluid continuity Eq. (9a) as

$$\nabla \cdot \mathbf{u} = -\phi_m(\rho_{np} - 1)[P\phi_t + \nabla \cdot (\phi\mathbf{u})]. \quad (24)$$

From here and based on the nanoparticle mass flux balance Eq. (9d), we obtain the expression for the divergence of the velocity vector

$$\nabla \cdot \mathbf{u} = -L(\rho_{np} - 1)[\nabla^2 \phi + \nabla \cdot (\eta\phi\nabla T)]. \quad (25)$$

The right-hand side of Eq. (25) represents the deviation of the system from an incompressible one. A possible simplification becomes feasible for moderate values of the average bulk particle concentration Φ in the case of a small Lewis number L , which ranges in $L \in (10^{-4}, 10^{-2})$, so that the parameter $L(\rho_{np} - 1) \in (10^{-4}, 10^{-2})$ for nanoparticles moderately heavier than the carrier fluid, and therefore the right - hand side of Eq. (25) may be neglected.

We now present a comparison between some results obtained for the problem at hand, governed in its full formulation by the equations with a non-uniform density referred to as “compressible” or “full” and those for its simplified version, where an incompressibility assumption is made by neglecting the right-hand side of Eq. (25) and retaining the rest of the governing equations and boundary condition with no change. This simplified formulation is referred to as an “incompressible” one. In some sense, the latter reminds the Boussinesq approximation in a nonisothermal fluid where the density is assumed to be constant except for allowing for the buoyancy force arising from inhomogeneity of the fluid density.

The left panel of Fig. 3 displays a comparison between the neutral curve $M_T(k)$ for a pure thermocapillary instability in the case of heating at the substrate for both ‘compressible’ and ‘incompressible’ formulations of the problem. We observe that the neutral curve of the simplified ‘incompressible’ problem exhibits an excellent agreement with the neutral curve obtained for the full ‘compressible’ formulation. The neutral curves $M_T(k)$ shown in the left panel of Fig. 3 for a variety of the Biot numbers B show that the instability is long-wave with a small but nonzero critical wavenumber $k = k_c$.

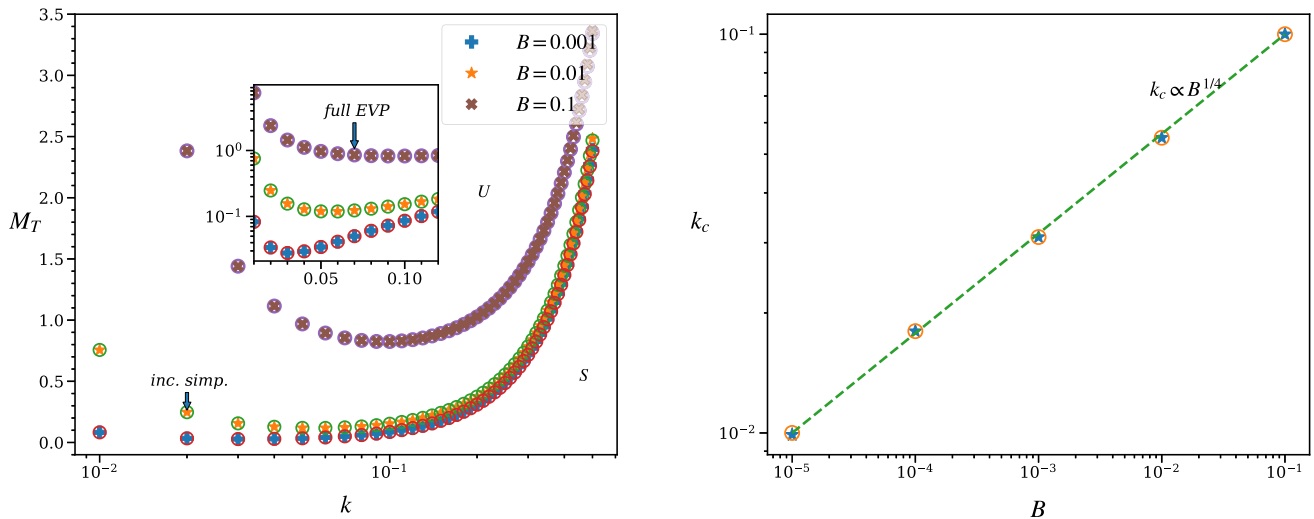
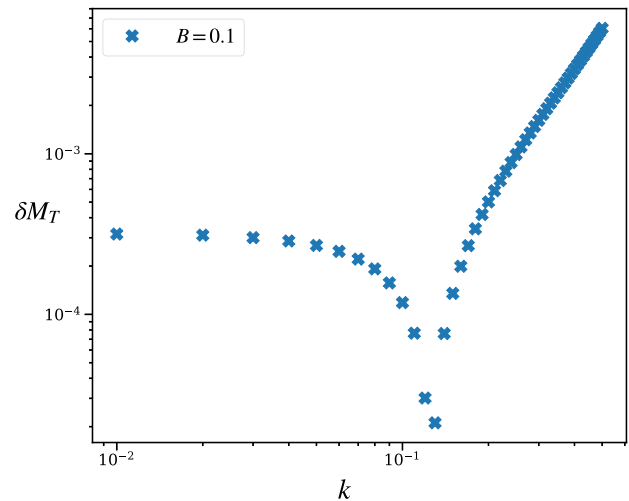


Fig. 3 Pure long-wave thermocapillary monotonic ($\lambda = 0$) instability in the case of heating at the substrate $Q = 1$ at $\Phi = 0.01$, $\eta = 0.31$, $L = 10^{-3}$, $a = 7.47$, $M_S = 0$, $\Sigma_0 = 10$, $G = 0.01$. Left panel: Neutral curves $M_T(k)$ for various Biot numbers B . S and U stand for the domains of stability and instability of the system, respectively. Right panel: Variation of the critical wavenumber k_c with the Biot number B . The dashed line represents the data fit $k_c \sim B^{1/4}$ with the factor of 0.1778. In both panels, circles ($\circ$) denote the results obtained from the numerical solution of the full EVP (23) for the corresponding values of B , whereas other symbols represent the results obtained from the numerical solution of the EVP in the “incompressible” formulation

Fig. 4 Variation of the relative error δM_T defined in Eq. (26) with the wavenumber k for a pure thermocapillary monotonic instability in the case of heating at the substrate $Q = 1$ for $B = 0.1$, $\Phi = 0.01$, $\eta = 0.31$, $L = 10^{-3}$, $a = 7.47$, $M_S = 0$, $\Sigma_0 = 10$, and $G = 0.01$



The right panel of Fig. 3 displays the variation of the critical wavenumber k_c with the Biot number B in the limit of $B \ll 1$. In this regime, the long-wave monotonic instability sets in and with an increase in the Biot number, the critical wavenumber k_c increases. In fact, the critical wavenumber obeys the $k_c \sim B^{1/4}$ scaling law which fits a similar result for a dilute binary mixture [14]. The results shown in this panel are obtained with and without the incompressibility simplification, and the agreement is excellent. Figure 4 shows the absolute value of the relative error between the full and incompressible formulations of the EVP (23) in terms of the thermal Marangoni number M_T . This relative error defined via

$$\delta M_T = \frac{|M_T - M_{T,inc}|}{|\max(M_T, M_{T,inc})|} \quad (26)$$

with M_T and $M_{T,inc}$ being, respectively, the values of M_T as obtained from solving the full and the incompressible formulations of the EVP (23), remains below 0.6%. We also note that near the critical wavenumber $k = k_c$, the difference δM_T between the two formulations is minimal, hence, we attain the minimal relative error around the threshold wavenumber. Therefore, we infer that the incompressible formulation may be safely used in our problem.

We reiterate that a use of the incompressible formulation reduces significantly the numerical effort required for solving the problem at hand.

5 Long-wave analysis in the limit $k_c \sim B^{1/4}$

Figure 3 displays the variation of the critical Marangoni number with the Biot number B in the limit of $B \ll 1$. As noted above, in this regime, the long-wave instability sets in and the instability follows the $k_c \sim B^{1/4}$ scaling law.

In what follows, we consider the case of a nanofluid layer with the deformable interface and introduce the following scaling with a small parameter ϵ into the eigenvalue problem Eq. (23)

$$B = \epsilon^4 \beta, \quad k = \epsilon K, \quad \lambda = \epsilon^2 \Lambda, \quad \Sigma_0 = O(1), \quad G = O(1). \quad (27)$$

Furthermore, we expand the perturbed field variables w , φ , θ , the growth rate λ and Marangoni numbers M_T and M_S into series of ϵ via

$$w = \epsilon^2(w_0 + \epsilon^2 w_2 + \dots), \quad (\theta, \varphi) = (\theta_0, \varphi_0) + \epsilon^2(\theta_2, \varphi_2) + \dots, \quad (28)$$

$$(\Lambda, M_T, M_S) = (\Lambda_0, M_{T0}, M_{S0}) + \epsilon^2(\Lambda_2, M_{T2}, M_{S2}) + \dots. \quad (29)$$

Substitution of the asymptotic expansions (27) and (29) into the eigenvalue problem (23) under the incompressibility assumption, see Sec. 4, results upon collecting the terms at the same order of ϵ in a sequence of linear boundary value problems.

At zero order, the energy and nanoparticle concentration equations read

$$-\frac{\eta \mathcal{P} \mathcal{Q} (\mathcal{Q} a \phi_m \varphi_0 + (\mathcal{P} + 1)^2 \theta'_0)}{(\mathcal{P} + 1)^3} + \frac{\mathcal{Q} a \phi_m \varphi'_0}{\mathcal{P} + 1} - (\mathcal{P} + 1) \theta''_0 = 0, \quad (30a)$$

$$\frac{\eta \left(\mathcal{Q} ((\mathcal{P} + 1)^2 \varphi'_0 - \eta \mathcal{P} \mathcal{Q} \varphi_0) - \frac{\mathcal{P} (\mathcal{P} + 1)^2 ((\mathcal{P} + 1) \theta''_0 + \eta \mathcal{Q} \theta'_0)}{a \phi_m} \right)}{(\mathcal{P} + 1)^3} - \varphi''_0 = 0, \quad (30b)$$

where $\mathcal{P} \equiv \mathcal{P}(z) = \mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \mathcal{Q} \eta z))$, which is associated with the Lambert function $\mathcal{W}$.

In the matrix form with the vector $\mathbf{A}_0 \equiv \mathbf{A}_0(z) = \begin{pmatrix} \theta_0 \\ \varphi_0 \end{pmatrix}$, Eqs. (30a)–(30b) are rewritten as

$$\mathcal{L} \mathbf{A}_0 = 0, \quad (30c)$$

where

$$\mathcal{L} = \begin{pmatrix} \left(\frac{-\mathcal{P} \mathcal{Q} \eta}{\mathcal{P} + 1} \right) \mathcal{D} - (1 + \mathcal{P}) \mathcal{D}^2 & -\frac{\eta \mathcal{P} \mathcal{Q}^2 a \phi_m}{(\mathcal{P} + 1)^3} + \left(\frac{\mathcal{Q} a \phi_m}{\mathcal{P} + 1} \right) \mathcal{D} \\ \left(-\frac{\eta^2 \mathcal{P} \mathcal{Q}}{(\mathcal{P} + 1) a \phi_m} \right) \mathcal{D} - \left(\frac{\eta \mathcal{P}}{a \phi_m} \right) \mathcal{D}^2 & -\frac{\eta^2 \mathcal{P} \mathcal{Q}^2}{(\mathcal{P} + 1)^3} + \left(\frac{\eta \mathcal{Q}}{\mathcal{P} + 1} \right) \mathcal{D} - \mathcal{D}^2 \end{pmatrix}, \quad (30d)$$

and $\mathcal{D} \equiv \frac{d}{dz}$. The corresponding homogeneous boundary conditions at zero order are

$$(\mathcal{P}(0) + 1) \mathcal{D} \theta_0 - \frac{\mathcal{Q} a \phi_m \varphi_0}{\mathcal{P}(0) + 1} = 0, \quad \frac{\eta \mathcal{P}(0) \mathcal{D} \theta_0}{a \phi_m} - \frac{\eta \mathcal{Q} \varphi_0}{\mathcal{P}(0) + 1} + \mathcal{D} \varphi_0 = 0 \quad \text{at } z = 0, \quad (30e)$$

$$(\mathcal{P}(1) + 1) \mathcal{D} \theta_0 - \frac{\mathcal{Q} a \phi_m \varphi_0}{\mathcal{P}(1) + 1} = 0, \quad \frac{\eta \mathcal{P}(1) \mathcal{D} \theta_0}{a \phi_m} - \frac{\eta \mathcal{Q} \varphi_0}{\mathcal{P}(1) + 1} + \mathcal{D} \varphi_0 = 0 \quad \text{at } z = 1. \quad (30f)$$

We note that Eqs. (30a)–(30f) representing a zero order approximation for the original set of equations and boundary conditions of the eigenvalue problem (23) do not contain the Biot number B which scales as $\mathcal{O}(\epsilon^4)$. This

results in the degeneracy of the problem at hand with an effective zero Biot number which in turn leads to the emergence of the solution $\theta_0^{(1)} = 1$, $\varphi_0^{(1)} = 0$ associated with a thermal mode.

There also exists a second linearly independent solution for Eqs. (30a)–(30f) associated with a slowly evolving concentration mode, which can be obtained by differentiation of the components of the steady base state of the system given by Eq. (17) with respect to the average volume concentration parameter $\gamma(\Phi)$. Therefore, a second solution for the zero order problem

$$\begin{aligned}\theta_0^{(2)} &= \left(\frac{dT_0}{d\gamma(\Phi)} \right) = \frac{(\gamma(\Phi)a\phi_m + 1)(\mathcal{P} - \mathcal{P}(1))}{\eta\gamma(\Phi)(\mathcal{P} + 1)(\mathcal{P}(1) + 1)}, \\ \varphi_0^{(2)} &= \left(\frac{d\phi_0}{d\gamma(\Phi)} \right) = \frac{(\gamma(\Phi)a\phi_m + 1)\mathcal{P}}{\gamma(\Phi)a\phi_m(\mathcal{P} + 1)}.\end{aligned}\quad (31)$$

We carry on with the solution for the zero order problem in the subspace spanned by its two nontrivial linearly independent solutions, namely,

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} \theta_0^{(2)} \\ \varphi_0^{(2)} \end{pmatrix}, \quad (32)$$

so the general solution of the problem (30a)–(30f) is given by

$$\theta_0 = \alpha_1\theta_0^{(1)} + \alpha_2\theta_0^{(2)}, \quad \varphi_0 = \alpha_1\varphi_0^{(1)} + \alpha_2\varphi_0^{(2)} \quad (33)$$

with two arbitrary constants α_1 and α_2 .

We now proceed to the derivation of the problem adjoint to Eqs. (30a)–(30f). Denote a vector $\mathbf{A}_0^* = \begin{pmatrix} \theta_0^* \\ \varphi_0^* \end{pmatrix}$ and a scalar product of two vectors $\mathbf{A}_0$ and $\mathbf{A}_0^*$ determined by $\langle \mathbf{A}_0^*, \mathbf{A}_0 \rangle = \int_0^1 dz(\theta_0^*\theta_0 + \varphi_0^*\varphi_0)$. The adjoint problem is derived by using integration by parts to obtain

$$\langle \mathbf{A}_0^*, \mathcal{L}\mathbf{A}_0 \rangle = \langle \mathcal{L}^\dagger \mathbf{A}_0^*, \mathbf{A}_0 \rangle + \mathcal{J}(\mathbf{A}_0, \mathbf{A}_0^*), \quad (34a)$$

where the linear adjoint operator is

$$\mathcal{L}^\dagger = \begin{pmatrix} -\left(\frac{\eta\mathcal{Q}\mathcal{P}}{\mathcal{P}+1}\right)\mathcal{D} - (1 + \mathcal{P})\mathcal{D}^2 & \left(-\frac{\eta^2\mathcal{Q}\mathcal{P}}{a\phi_m(\mathcal{P}+1)}\right)\mathcal{D} - \left(\frac{\eta\mathcal{P}}{a\phi_m}\right)\mathcal{D}^2 \\ -\left(\frac{\mathcal{Q}a\phi_m}{\mathcal{P}+1}\right)\mathcal{D} & -\left(\frac{\eta\mathcal{Q}}{\mathcal{P}+1}\right)\mathcal{D} - \mathcal{D}^2 \end{pmatrix}, \quad (34b)$$

and the boundary conditions obtained from the condition of a zero bilinear concomitant $\mathcal{J}(\mathbf{A}_0, \mathbf{A}_0^*)$

$$\frac{\eta\mathcal{P}(0)\varphi_0^{*'}}{a\phi_m} + (\mathcal{P}(0) + 1)\theta_0^{*'} = 0, \quad \varphi_0^{*'} = 0 \quad \text{at } z = 0, \quad (34c)$$

$$\frac{\eta\mathcal{P}(1)\varphi_0^{*'}}{a\phi_m} + (\mathcal{P}(1) + 1)\theta_0^{*'} = 0, \quad \varphi_0^{*'} = 0 \quad \text{at } z = 1. \quad (34d)$$

The adjoint problem

$$\mathcal{L}^\dagger \mathbf{A}_0^* = 0 \quad (35)$$

has been simplified to the form presented in Eq. (34b) using the relationship

$$\mathcal{D}\mathcal{P} = \frac{\eta\mathcal{Q}\mathcal{P}}{1 + \mathcal{P}}, \quad (36)$$

which can be easily derived via the inverse function theorem. By virtue of this property of the Lambert function $\mathcal{W}$, all entries of $\mathcal{L}^\dagger$ not containing the differentiation operator $\mathcal{D}$ cancel out. More details are given in Appendix A. We note in passing that $\mathcal{L} \neq \mathcal{L}^\dagger$, which implies that the problem given by Eq. (30) is not self-adjoint.

We now determine the eigenvectors of the adjoint problem given by Eqs. (35), (34c), (34d) corresponding to the zero eigenvalue of $\mathcal{L}^\dagger$. We find that there exist two eigenvectors and we find them analytically in the case of heating at the substrate $\mathcal{Q} = 1$

$$\theta_0^* = c_1 z + c_2, \quad \varphi_0^* = c_3 \left(\ln(|\mathcal{P}|) - \frac{1}{\mathcal{P}} \right) - \frac{c_1 a \phi_m}{\eta} z + c_4, \quad (37)$$

where $c_1, \dots, c_4$ are constants, which are determined by substitution into the boundary conditions Eqs. (34c) and (34d) resulting in $c_1 = c_3 = 0$. Therefore, the zero eigenvectors of the adjoint problem (34) are $\theta_0^* = c_2$ and $\varphi_0^* = c_4$, where the constants c_2, c_4 are arbitrary. Consequently, the linearly independent eigenvectors of the adjoint operator corresponding to its zero eigenvalue are

$$\mathbf{u}_1^* = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{u}_2^* = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (38)$$

We return to the momentum conservation equation at zero order and the corresponding boundary conditions that read

$$GK^2 \phi_m (\rho_{np} - 1) (\mathcal{P} + 1)^2 (\mathcal{P} - a\phi_m)^3 \varphi_0 + \frac{2\eta^2 \mathcal{Q}^2 (a\phi_m)^2 \mathcal{P} [a\phi_m + \mathcal{P}(3\mathcal{P} + 2)]}{(\mathcal{P} + 1)(\mathcal{P} - a\phi_m)} \mathcal{D}^2 w_0 \\ + (\mathcal{P} + 1)(a\phi_m)^2 [(\mathcal{P} + 1)(\mathcal{P} - a\phi_m) \mathcal{D}^4 w_0 - 4\eta \mathcal{Q} \mathcal{P} \mathcal{D}^3 w_0] = 0, \quad (39a)$$

$$w_0 = \mathcal{D}w_0 = 0 \quad \text{at } z = 0, \quad (39b)$$

At $z = 1$:

$$\zeta_0 \Lambda_0 P - w_0 = 0, \quad (39c)$$

$$\frac{\eta \zeta_0 K^2 \mathcal{P}(1) \mathcal{Q} M_{S0}}{(\mathcal{P}(1) + 1) a \phi_m} + \frac{(a\phi_m)^2 \mathcal{D}^2 w_0}{(a\phi_m - \mathcal{P}(1))^2} + K^2 \left[M_{T0} \left(\theta_0 - \frac{\zeta_0 \mathcal{Q}}{\mathcal{P}(1) + 1} \right) + M_{S0} \varphi_0 \right] = 0, \quad (39d)$$

with

$$\zeta_0 = \frac{(a\phi_m)^3 [(\mathcal{P}(1) + 1)(a\phi_m - \mathcal{P}(1)) \mathcal{D}^3 w_0 + 2\eta \mathcal{P}(1) \mathcal{Q} \mathcal{D}^2 w_0]}{GK^2 (\mathcal{P}(1) + 1)(a\phi_m - \mathcal{P}(1))^3 [a\phi_m + \mathcal{P}(1) \phi_m (\rho_{np} - 1)]}. \quad (39e)$$

The energy and nanoparticle concentration equations with corresponding boundary conditions at order ϵ^2 reads

$$- \frac{\eta \mathcal{Q} \mathcal{P} (\mathcal{Q} a \phi_m \varphi_2 + (\mathcal{P} + 1)^2 \mathcal{D} \theta_2)}{(\mathcal{P} + 1)^3} + \frac{\mathcal{Q} a \phi_m \mathcal{D} \varphi_2}{\mathcal{P} + 1} - (\mathcal{P} + 1) \mathcal{D}^2 \theta_2 \\ = - \frac{\theta_0 [a(K^2 (\mathcal{P} + 1) + \Lambda_0 P) + \Lambda_0 P ((\rho c)_{np} - 1) \mathcal{P}]}{a} + \frac{\mathcal{Q} [a + \mathcal{P} ((\rho c)_{np} - 1)] w_0}{(\mathcal{P} + 1) a}, \quad (40a)$$

$$- \frac{\eta \mathcal{P} (\eta \mathcal{Q} \mathcal{D} \theta_2 + (\mathcal{P} + 1) \mathcal{D}^2 \theta_2)}{(\mathcal{P} + 1) a \phi_m} + \frac{\eta \mathcal{Q} [(\mathcal{P} + 1)^2 \mathcal{D} \varphi_2 - \eta \mathcal{P} \mathcal{Q} \varphi_2]}{(\mathcal{P} + 1)^3} - \mathcal{D}^2 \varphi_2 \\ = - \frac{\eta K^2 \mathcal{P} \theta_0}{a \phi_m} - \frac{\eta \mathcal{Q} \mathcal{P} w_0}{L a \phi_m (\mathcal{P} + 1)} - \frac{\varphi_0 (K^2 L + \Lambda_0 P)}{L}, \quad (40b)$$

$$(\mathcal{P}(0) + 1) \mathcal{D} \theta_2 - \frac{\mathcal{Q} a \phi_m \varphi_2}{\mathcal{P}(0) + 1} = 0, \quad \frac{\eta \mathcal{P}(0) \mathcal{D} \theta_2}{a \phi_m} - \frac{\eta \mathcal{Q} \varphi_2}{\mathcal{P}(0) + 1} + \mathcal{D} \varphi_2 = 0 \quad \text{at } z = 0, \quad (40c)$$

$$(\mathcal{P}(1) + 1) \mathcal{D} \theta_2 - \frac{\mathcal{Q} a \phi_m \varphi_2}{\mathcal{P}(1) + 1} = 0, \quad \frac{\eta \mathcal{P}(1) \mathcal{D} \theta_2}{a \phi_m} - \frac{\eta \mathcal{Q} \varphi_2}{\mathcal{P}(1) + 1} + \mathcal{D} \varphi_2 = 0 \quad \text{at } z = 1. \quad (40d)$$

In what follows, we concentrate on finding the threshold for the monotonic instability, $\Lambda_0 = 0$, and rewrite the second-order Eqs. (40a) and (40b) in the form

$$\mathcal{L}A_2 = R, \quad (41a)$$

$$R = - \left(\begin{array}{c} K^2(\mathcal{P} + 1)\theta_0 - \left(\frac{\mathcal{Q}(a + ((\rho\mathcal{C})_{np} - 1)\mathcal{P})}{a(\mathcal{P} + 1)} \right) w_0(M_{T0}, M_{S0}) \\ \frac{\eta\mathcal{P}[K^2L(\mathcal{P} + 1)\theta_0 + \mathcal{Q}w_0(M_{T0}, M_{S0})]}{La\phi_m(\mathcal{P} + 1)} + K^2\varphi_0 \end{array} \right), \quad (41b)$$

where $A_2 = \begin{pmatrix} \theta_2 \\ \varphi_2 \end{pmatrix}$.

The sets of linear equations (30c) and (41a) possess the same matrix $\mathcal{L}$ while the former is homogeneous and the latter is nonhomogeneous. Therefore, we invoke the Fredholm alternative theorem [14, 33–38] to impose the solvability condition that in the case at hand, consists of two conditions of orthogonality between the vector R and the eigenvectors of the adjoint operator u_1^* and u_2^*

$$\langle R, u_1^* \rangle = 0, \quad \langle R, u_2^* \rangle = 0, \quad (42)$$

which turn out to be equivalent to just a separate integration of the components of the vector R across the layer and equating its results to zero.

The problem (39e) is solved numerically in terms of w_0 with $\begin{pmatrix} \theta_0 \\ \varphi_0 \end{pmatrix}$ taken alternately as u_1 and u_2 with the parameters M_{T0} and M_{S0} for fixed G , η , a , ϕ_m , ρ_{np} , and $\mathcal{Q} = 1$ resulting in the solutions $w_0^{(1)}(M_{T0}, M_{S0})$ and $w_0^{(2)}(M_{T0}, M_{S0})$, respectively. Thus, the general solution for $w_0(M_{T0}, M_{S0})$ is $w_0(M_{T0}, M_{S0}) = \alpha_1 w_0^{(1)}(M_{T0}, M_{S0}) + \alpha_2 w_0^{(2)}(M_{T0}, M_{S0})$.

The solvability conditions (42) are recast as

$$\left\{ \begin{array}{l} \int_0^1 \left[K^2(\mathcal{P} + 1)\theta_0 - \left(\frac{\mathcal{Q}(a + ((\rho\mathcal{C})_{np} - 1)\mathcal{P})}{a(\mathcal{P} + 1)} \right) w_0(M_{T0}, M_{S0}) \right] dz = 0, \\ \int_0^1 \left[\frac{\eta\mathcal{P}[K^2L(\mathcal{P} + 1)\theta_0 + \mathcal{Q}w_0(M_{T0}, M_{S0})]}{La\phi_m(\mathcal{P} + 1)} + K^2\varphi_0 \right] dz = 0, \end{array} \right. \quad (43)$$

with θ_0 and φ_0 given from Eq. (33). Substituting θ_0 , φ_0 and $w_0(M_{T0}, M_{S0})$ into Eq. (43) results in a linear set of two homogeneous equations with respect to the unknowns α_1 and α_2 . Since $w_0(M_{T0}, M_{S0})$ depends linearly on both M_{T0} and M_{S0} via the boundary condition (39d), all the four entries of the resulting matrix of this set of equations are linear functions of M_{T0} and M_{S0} . The critical value of M_{T0} will be determined for a fixed but varying M_{S0} and imposing a zero value requirement to the matrix determinant resulting in a quadratic equation in terms of M_{T0} .

It is important to note that w_0 as determined by Eqs. (39a) and (39b)-(39d) at $\Lambda_0 = 0$ is proportional to K^2 and ζ_0 as given by Eq. (39e) is independent of K . Consequently, the solutions of Eq. (43) in terms of M_{T0} and M_{S0} are independent of the wavenumber K .

Recently, Gandhi et al. [20] have revealed various transitions from a monotonic thermocapillary instability to an oscillatory one for the system at hand with either deformable or nondeformable interface. The left panel of Fig. 5 displays the variation of the critical value of the thermal Marangoni number M_T with the Biot number B . We note that in the limit of $k \sim B^{1/4}$ considered in Sect. 5, the monotonic long-wave pure thermocapillary instability takes place and its critical value of M_T is represented by the $\times$ point in the left panel of Fig. 5. We emphasize that our current long-wave theory is evaluated up to the second order, whereas the Biot number contributes in the fourth order of approximation. Thus, the effective Biot number in our current approximation for the critical thermal Marangoni number is zero. Consequently, the stabilizing effect of the Biot number B is shown in Fig. 5 via the numerical solution of the EVP (23). Notably, we observe here a switch from the long-wave monotonic thermocapillary instability to the long-wave oscillatory thermocapillary instability for a small but non-zero Biot number. Interestingly, in [20] we stipulated the emergence of an oscillatory thermocapillary instability due to variation of the thermal conductivity stratification parameter a , for more details, see [20]. We conjecture that the inherent mechanism responsible for the onset of the oscillatory instability is due to an apparent competition between the destabilizing thermocapillarity weakening with an increase in the Biot number and the stabilization

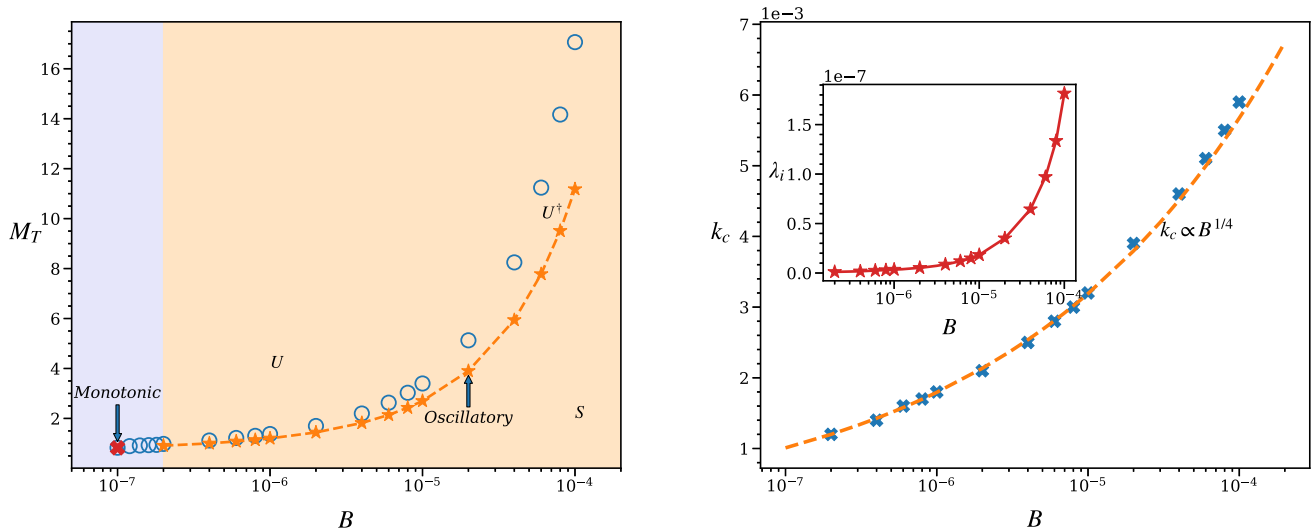


Fig. 5 Left panel: Variation of the threshold values of the Marangoni number M_T in the case of a pure thermocapillary instability with the Biot number B in the case of heating at the substrate $\mathcal{Q} = 1$ for $\Phi = 0.01$, $\eta = 0.31$, $G = 1$, $M_S = 0$, and $\Sigma_0 = 10^5$. Here, $\circ$, $\star$, and $\times$ points denote the thresholds of the monotonic and oscillatory instabilities, both found numerically by solving the incompressible version of the EVP (23), and the long-wave monotonic thermocapillary instability found using the results of Sect. 5, respectively. The symbols U , S , and $U^\dagger$ stand for the domains of monotonic instability, stability, and oscillatory instability, respectively. Right panel: Variation of the critical wavenumber k_c of the oscillatory branch of instability for the case shown in the left panel with the Biot number B ; Inset right panel: Variation of the critical frequency $\lambda_i \equiv \Im(\lambda)$ with B ; The dashed line represents the data fit $k_c \sim B^{1/4}$ with the proportionality factor of 0.0567 for the oscillatory instability

mechanism of thermal diffusion induced by the thermal stratification parameter a and the Soret effect. In addition, the right panel of Fig. 5 displays the variation of the critical wavenumber k_c of the long-wave oscillatory branch with B . We note that k_c follows the well-known characteristic scaling law $k_c \sim B^{1/4}$ with the proportionality constant being ≈ 0.057 . We also note that the critical frequency $\Im(\lambda)$ varies monotonically with the Biot number, $\lambda_i \sim B^{3/4}$, as shown in the inset to the right panel of Fig. 5.

Let us emphasize that the oscillatory instability appears even for $M_S = 0$. Therefore, the mechanism of the oscillatory instability is different from that considered by Joo [13] and Oron and Nepomnyashchy [14] for dilute binary mixtures, where it arises as a result of the competition between the destabilizing thermocapillarity and the stabilizing solutocapillarity. Here, the role of a stabilizing mode is played by the nanoparticle diffusion mode. The oscillations develop due to an apparent disparity between the thermal and nanoparticle diffusional time scales, i.e., $L \ll 1$. Figure 6 illustrates the variation of the threshold values of the thermal Marangoni number M_T with the solutal Marangoni number M_S . We observe a monotonic increase in the critical M_T with an increase in M_S and find a good agreement between the results of the numerical solution based on the incompressible version of Eq. (23) and the long-wave theory based on Sect. 5. In addition, we note that both the critical thermal Marangoni number M_T along the oscillatory branch and the critical frequency λ_i vary weakly with the solutal Marangoni number M_S . We also note that the critical wavenumber k_c in the monotonic mode remains around $k_c \approx 4 \times 10^{-4}$, however, a discontinuity in k_c takes place at the origin of the oscillatory mode with $k_c \approx 10^{-3}$. This suggests that the transition from the monotonic to oscillatory thermocapillary instability occurs via a codimension-two point.

We also find that in terms of the critical Marangoni number M_T our long-wave theory converges to the reference case of monotonic thermocapillary instability in a pure liquid $M = \frac{2}{3}G$ [6, 9] when the Soret parameter, thermal conductivity stratification parameter a and average bulk concentration are vanishingly small. This implies that this convergence is irrespective of the value of dimensionless surface tension Σ_0 .

Figure 7 displays the variation of the critical value of M_T in the case of a pure long-wave thermocapillary instability with the thermal conductivity parameter a for various values of the average particle concentration Φ when the layer interface is nondeformable and the Soret coefficient η is negligibly small. We find that the critical value of M_T monotonically increases with a for a fixed Φ and with Φ when a is fixed. The former is associated with an increased viscosity as a result of a higher particle concentration. It is also observed in Figure 7 that with a sufficiently low value of the thermal conductivity stratification parameter a and the average particle concentration Φ the critical value of M_T tends to the value $M_T = 48$ corresponding to the classical result given by Pearson [8]. The results found using the asymptotic expansion method in the limit of Sect. 5 agree very well with those found

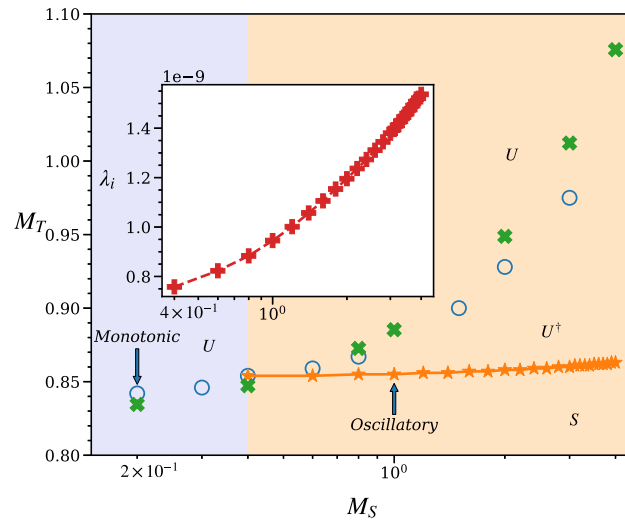


Fig. 6 Variation of the threshold values of the thermal Marangoni number M_T with the solutal Marangoni number M_S in the case of heating at the substrate $\mathcal{Q} = 1$ for $\Phi = 0.01$, $\eta = 0.31$, $G = 1$, $B = 10^{-7}$, and $\Sigma_0 = 10^5$; Inset: Variation of the critical frequency $\lambda_i \equiv \Im(\lambda)$ with M_S along the critical curve in the oscillatory domain. Here, $\circ$, $\star$, and $\times$ points denote the thresholds of the monotonic and oscillatory instabilities, both obtained by numerically solving the incompressible version of the EVP (23), and the long-wave monotonic instability obtained using the asymptotic expansions, Sect. 5, respectively. The symbols U , S , and U^+ stand for the domains of monotonic instability, stability, and oscillatory instability of the system, respectively

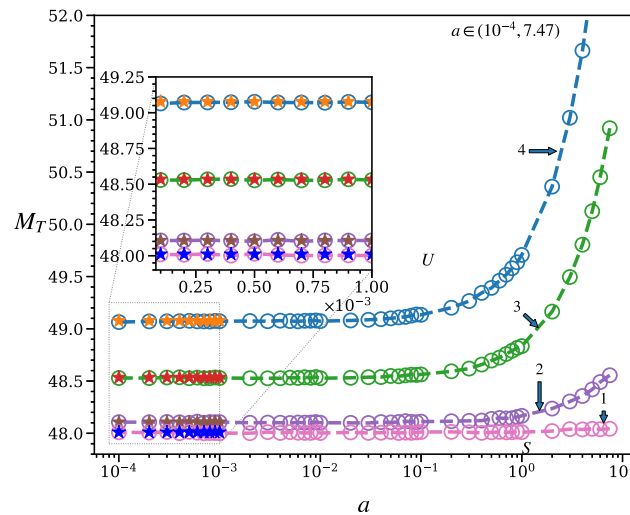


Fig. 7 Variation of the threshold of the pure long-wave thermocapillary monotonic instability with the thermal conductivity stratification parameter a in the case of heating at the substrate $\mathcal{Q} = 1$ with $\eta = 10^{-8}$, $L = 10^{-3}$, $k \rightarrow 0$, $M_S = 0$, $\zeta_0 = 0$, and different values of the average bulk nanoparticle concentration Φ . The $\circ$ and $\star$ points represent the values obtained via full numerical solution of the incompressible version of the eigenvalue problem (23) and using our asymptotic long-wave approach in the limit of Sect. 5, respectively. S and U denote the domains of stability and instability of the system, respectively. The inset shows the results in the domain of $10^{-4} \leq a \leq 10^{-3}$. Curves 4 to 1 represent the average bulk nanoparticle concentration $\Phi = 10^{-2}$, 5×10^{-3} , 10^{-3} , and 10^{-4} , respectively

via numerical solution of the incompressible formulation of the EVP (23) as seen in the inset for the entire interval of the monotonic instability in terms of a .

6 Summary and conclusions

In this paper, we carry out the long-wave linear asymptotic stability analysis of the monotonic thermosolutocapillary instability coupled with the Soret effect and driven by a specified heat flux at the solid substrate in a layer of a moderately dense nanofluid. The mathematical model for such nanofluid layers with a deformable interface exposed to the gas phase in a gravity field was presented in [20]. The dynamics of the considered system is governed by the continuity, momentum, energy, and nanoparticle volume conservation equations. All thermophysical properties of the nanofluid such as its density, viscosity, heat capacity, thermal conductivity, and thermodiffusion coefficient are assumed to depend on the local particle volume concentration.

In the framework of our model, we find a steady-state quiescent base state expressed by the analytical function known as the Lambert W function. We note that the base-state of the system when heated at the substrate exhibits a nonlinear variation of temperature and particle concentration fields with the nanofluid height at moderate-temperature differences across the layer. Linear stability analysis of this base state of the system in the case of the monotonic mode is performed here in the long-wave limit and its results are compared with those of the full linear stability investigation [20].

We also find that due to a small value of the Lewis number, it is possible to simplify the continuity equation and to bring it to the form which is identical to that valid for incompressible fluids. A comparison between the neutral curves for the critical thermal Marangoni number obtained for varying values of the Biot number obtained by solving the full ('compressible') eigenvalue problem and its simplified 'incompressible' version differ by less than 0.6 %.

Based on this observation, in this paper, we use the simplified version of the problem for all our derivations. Gandhi et al. [20] found that both monotonic and oscillatory instabilities may emerge in nanofluids with the finite critical wavenumber. However, with a decrease in the Biot number B , the critical wavenumber k_c tends to zero as $k_c \sim B^{1/4}$. Among these cases, there exist parameter regimes where the long-wave monotonic instabilities set in. This is where are long-wave asymptotic derivations are valid.

Our long-wave analysis yields the critical values for the thermal and solutal Marangoni numbers via solvability conditions at second order of expansion. Unfortunately, we have been unsuccessful in solving analytically the second-order problem, primarily due to the presence of w_0 in the right-hand side of the problem which can be only evaluated numerically, to employ the solvability condition at fourth-order of approximation to find the second-order corrections for the critical Marangoni number and also the leading order term for the critical wavenumber. Hence, at the current order of approximation, the critical wavenumber is zero, and to ascertain that the values of both Marangoni numbers are indeed critical, we rely on the comparison between the results of our asymptotic theory and those obtained in the fully numerical solution of the simplified 'incompressible' version of the general eigenvalue problem (23) which is found to be excellent.

It is important to emphasize that numerical solution of the EVP (23) in either its full or incompressible versions involves a significant computational effort. On the other hand, the long-wave analysis presented here in the case of the monotonic instability predicting the critical values of the control parameter such as the thermal and/or solutal Marangoni numbers is analytical accompanied with numerical solution of an ordinary differential equation and therefore is not computationally expensive.

Finally, our numerical analysis of the problem in both full and simplified 'incompressible' versions shows also the emergence of long-wave oscillatory instabilities. Therefore, an extension of our long-wave analysis to determine the threshold of oscillatory instability is needed to complete the investigation of the stability properties of the system at hand. In fact, this extension is now in progress and the results will be published elsewhere.

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Appendix A. Adjoint problem operator

The linear operator $\mathcal{L}$ of Eq. (30d) is rewritten in the short form as

$$\mathcal{L} = \begin{pmatrix} f_1\mathcal{D} + f_2\mathcal{D}^2 & f_3 + f_4\mathcal{D} \\ f_5\mathcal{D} + f_6\mathcal{D}^2 & f_7 + f_8\mathcal{D} - \mathcal{D}^2 \end{pmatrix}, \quad (44)$$

where

$$\begin{aligned}
f_1 &= -\frac{\eta \mathcal{P} \mathcal{Q}}{\mathcal{P} + 1}, & f_2 &= -(1 + \mathcal{P}), & f_3 &= -\frac{\eta \mathcal{P} \mathcal{Q}^2 a \phi_m}{(\mathcal{P} + 1)^3}, & f_4 &= \frac{\mathcal{Q} a \phi_m}{\mathcal{P} + 1}, \\
f_5 &= -\frac{\eta^2 \mathcal{P} \mathcal{Q}}{(\mathcal{P} + 1) a \phi_m}, & f_6 &= -\frac{\eta \mathcal{P}}{a \phi_m}, & f_7 &= -\frac{\eta^2 \mathcal{P} \mathcal{Q}^2}{(\mathcal{P} + 1)^3}, & f_8 &= \frac{\eta \mathcal{Q}}{\mathcal{P} + 1}
\end{aligned} \tag{45}$$

are functions of z containing the function

$$\mathcal{P} \equiv \mathcal{P}(z) = \mathcal{W}(\gamma(\Phi) a \phi_m \exp(\gamma(\Phi) a \phi_m + \mathcal{Q} \eta z)). \tag{46}$$

We note that the left-hand side of the zero order problem may be presented in the factorized form

$$\mathcal{L} \mathbf{A}_0 = \begin{pmatrix} \mathcal{D} & \mathcal{D} \\ \mathcal{D} & \mathcal{D} \end{pmatrix} \left[\begin{pmatrix} f_2 & f_4 \\ f_6 & f_8 - \mathcal{D} \end{pmatrix} \begin{bmatrix} \mathcal{D} \theta_0 \\ \varphi_0 \end{bmatrix} \right]. \tag{47}$$

The adjoint linear operator is obtained in the form of

$$\mathcal{L}^\dagger = \begin{pmatrix} (-\mathcal{D} f_1 + \mathcal{D}^2 f_2) + (-f_1 + 2\mathcal{D} f_2) \mathcal{D} + f_2 \mathcal{D}^2 & (-\mathcal{D} f_5 + \mathcal{D}^2 f_6) + (2\mathcal{D} f_6 - f_5) \mathcal{D} + f_6 \mathcal{D}^2 \\ (f_3 - \mathcal{D} f_4) - f_4 \mathcal{D} & (f_7 - \mathcal{D} f_8) - f_8 \mathcal{D} - \mathcal{D}^2 \end{pmatrix}. \tag{48}$$

It is easily checked that $\mathcal{D} f_2 = f_1$, $\mathcal{D} f_6 = f_5$, $f_3 = \mathcal{D} f_4$, and $f_7 = \mathcal{D} f_8$, and therefore the left-hand side of the adjoint problem may be written in the factorized form as

$$\mathcal{L}^\dagger \mathbf{A}_0^* = \begin{pmatrix} f_1 + f_2 \mathcal{D} & f_6 \mathcal{D} + \mathcal{D} f_6 \\ -f_4 & -f_8 - \mathcal{D} \end{pmatrix} \begin{pmatrix} \mathcal{D} & 0 \\ 0 & \mathcal{D} \end{pmatrix} \cdot \begin{pmatrix} \theta_0^* \\ \varphi_0^* \end{pmatrix}. \tag{49}$$

Appendix B. Solution for the adjoint problem (35)–(34b)

We begin the solution procedure by introducing new variables $u = \mathcal{D} \theta_0^*$ and $v = \mathcal{D} \varphi_0^*$. In terms of the new variables, the zero-order equations read

$$\left(\frac{\eta \mathcal{Q} \mathcal{P}}{\mathcal{P} + 1} \right) u + (1 + \mathcal{P}) \mathcal{D} u + \left(\frac{\eta^2 \mathcal{Q} \mathcal{P}}{a \phi_m (\mathcal{P} + 1)} \right) v + \left(\frac{\eta \mathcal{P}}{a \phi_m} \right) \mathcal{D} v = 0, \tag{50a}$$

$$\left(\frac{\mathcal{Q} a \phi_m}{\mathcal{P} + 1} \right) u + \left(\frac{\eta \mathcal{Q}}{\mathcal{P} + 1} \right) v + \mathcal{D} v = 0. \tag{50b}$$

Eliminating u from Eq. (50b) and substituting it into Eq. (50a) yields $u = c_1$ and therefore $\theta_0^* = c_1 z + c_2$, where c_1, c_2 are constants to be found from the boundary conditions (34c)–(34d).

Substituting $u = c_1$ into Eq. (50a) results in

$$\mathcal{D} \varphi_0^* \equiv v = -\frac{a \phi_m c_1}{\eta} + c_3 \frac{1}{\mathcal{P} \mathcal{Q}}, \tag{51}$$

where c_3 is a constant.

It is possible to solve Eq. (51) in terms of φ_0^* for both cases of heating and cooling at the substrate. In the case of heating at the substrate $\mathcal{Q} = 1$, using the property of $\mathcal{P}$ expressed by Eq. (36), the solution for φ_0^* is obtained as

$$\varphi_0^* = c_3 \left(\ln(|\mathcal{P}|) - \frac{1}{\mathcal{P}} \right) - \frac{c_1 a \phi_m}{\eta} z + c_4, \tag{52}$$

with c_4 being an arbitrary constant. Therefore, the solution of the adjoint problem corresponding to its zero eigenvalue is

$$\theta_0^* = c_1 z + c_2, \quad \varphi_0^* = c_3 \left(\ln(|\mathcal{P}|) - \frac{1}{\mathcal{P}} \right) - \frac{c_1 a \phi_m}{\eta} z + c_4, \quad (53)$$

where the constant c_i , $i = 1, \dots, 4$ are to be found from the boundary conditions (34c)-(34d).

We note in passing that in the case of cooling at the substrate, $\mathcal{Q} = -1$, applying again Eq. (36), Eq. (50a) becomes

$$\begin{aligned} \mathcal{D}\varphi_0^* &= -\frac{a\phi_m c_1}{\eta} + c_3 \mathcal{P} \\ &= -\frac{a\phi_m c_1}{\eta} - \frac{c_3}{\eta} (\mathcal{P} + 1) \mathcal{D}\mathcal{P}. \end{aligned} \quad (54)$$

Therefore, the solution for φ_0^* and θ_0^* is obtained as

$$\theta_0^* = c_1 z + c_2, \quad \varphi_0^* = -\frac{a\phi_m c_1}{\eta} z + c_4 - \frac{c_3}{2\eta} (\mathcal{P} + 1)^2. \quad (55)$$

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